

ARXIV.ORG/ABS/2504.04715 · LLM API INTEGRITY

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs: why software tests fail and hardware attestation wins.

Trust gap: black-box APIs cannot prove the advertised model is served



THREAT MODEL

The Model Substitution Problem

H_0

$$P_{\text{actual}}(y|x) = P_{\text{spec}}(y|x)$$

H_1

$$P_{\text{actual}}(y|x) \neq P_{\text{spec}}(y|x)$$

Given API samples, the auditor must distinguish the specified model distribution from an alternative distribution.

Economic incentive: reduce GPU costs while maintaining pricing.

Smaller model

Example: serving an 8B model instead of a 70B model.

Fine-tuned update

A tuned or revised model behind the same API name.

Quantized variant

FP8 or INT8 instead of original precision.

Different family

An entirely different architecture or provider lineage.

API providers charge for specific models, but black-box serving gives users no direct guarantee that the advertised model is actually executed.

EMPIRICAL EVIDENCE

FP8 Quantization Preserves Most Accuracy

Across five model families, benchmark differences are small enough that substitution becomes hard to separate from ordinary measurement variation.

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
Llama-3-8B-Instruct-FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
Llama-3-70B-Instruct-FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
Gemma-2-9b-it-FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
Qwen2-72B-Instruct-FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
Mistral-7B-Instruct-v0.3-FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

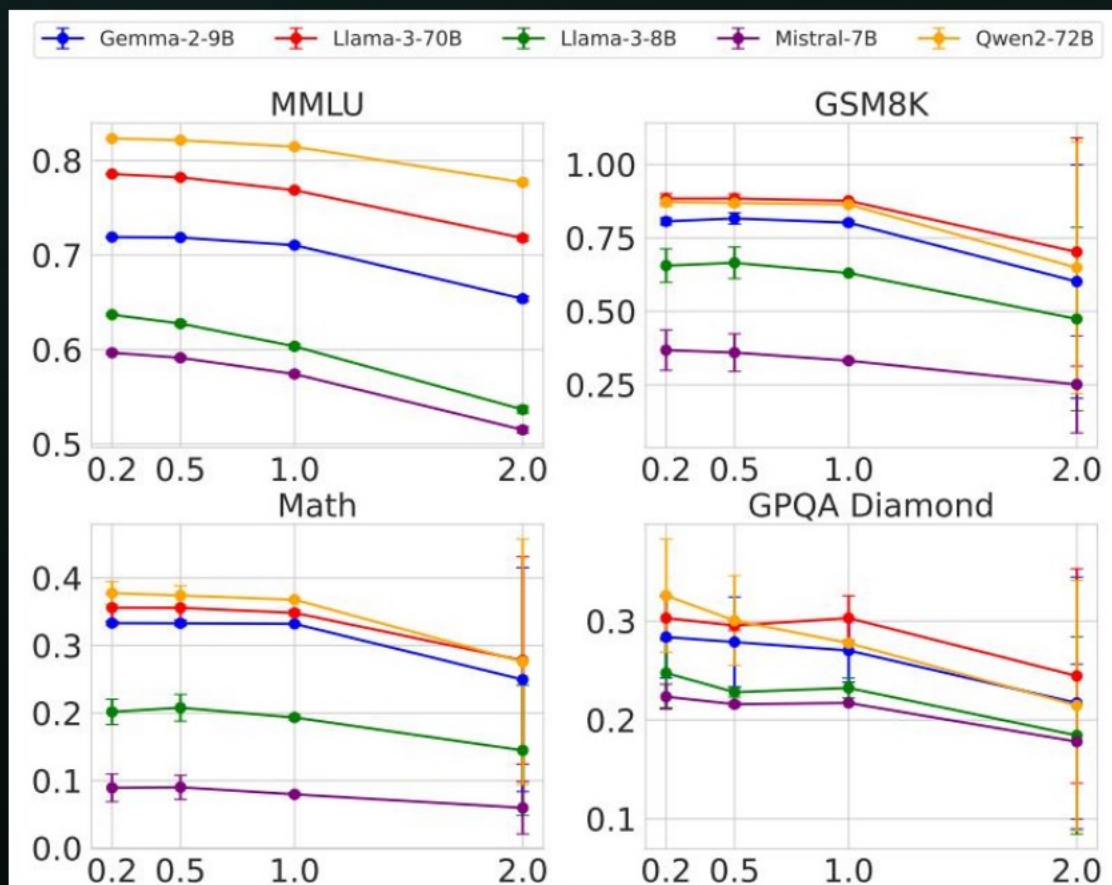
FP8 rows shaded

Many drops fall within error bars

MMLU and GSM8K especially subtle

Higher Temperature Amplifies Gaps but Adds Noise

Temperature experiments at 0.2, 0.5, 1.0, and 2.0 show performance degradation across models, making substitution signals less stable.



Across-the-board degradation

At temperature 2.0, performance degrades broadly rather than isolating a clean substitute signature.

Noisy separability

The original/substitute gap can narrow or become noisy as sampling temperature increases.

Operational implication

Auditors cannot assume benchmark accuracy is a stable fingerprint under realistic serving settings.

Log Probability Fingerprinting Fails

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

The same Llama-3-8B model can emit different log probabilities across inference frameworks, versions, and GPU hardware.

Framework drift

Variation appears across transformers 4.40 vs 4.50 and vLLM 0.6.2 vs 0.8.2.

Hardware drift

A100 and H100 deployments introduce production-level differences.

Audit failure

An auditor cannot distinguish legitimate infrastructure variation from actual substitution.

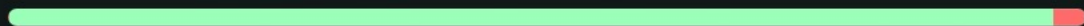
Low-Rate Randomization Becomes Query-Prohibitive

A provider can route only a fraction of traffic to the cheaper model. As that fraction approaches zero, the mixed distribution converges toward the specified model.

$$P_{\text{mix}} = (1 - p)P_{\text{spec}} + pP_{\text{sub}}$$

Most queries

Served by the specified model, preserving the expected output distribution.

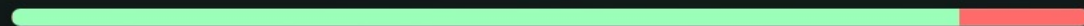


p approaches 0



Small hidden fraction

Routed to a cheaper substitute, reducing cost while diluting detection power.



mixed distribution approaches P_{spec}

Detection requires many more queries

Benchmark evasion can route known prompts to the genuine model

FAILURE MATRIX

Software-Only Auditing Is Unreliable

Text outputs, metadata, and benchmarks all depend on distributions that can be noisy, subtle, or adversarially shaped.

Method type	Weakness	Practical limitation
Text-based statistical tests Examples include MMD over generated text.	Subtle substitutions and FP8 variants barely shift benchmark-visible behavior.	Require impractically large query budgets, especially under randomized routing.
Log-probability methods Compare token-level metadata to a local reference.	Production nondeterminism changes log probabilities across software and hardware.	Cannot separate legitimate deployment differences from model substitution.
Adversarial countermeasures Randomized substitution and benchmark evasion.	The provider controls routing and can recognize known benchmark prompts.	Software-only detection becomes fundamentally query-prohibitive.

ACTIONABLE ALTERNATIVE

TEEs Provide Cryptographic Model Integrity

Trusted Execution Environments shift verification below the software audit layer: attest the code and data actually loaded for inference.

1 • Enclave

Run inference inside a hardware-secured environment.



2 • Weights

Verify specific model weights are loaded.



3 • Code

Verify the correct inference code is executing.



4 • Attestation

Return cryptographic evidence to the client.

Provable

Integrity is hardware-attested, not inferred statistically.

Efficient

The brief reports only modest performance overhead.

Deployable

NVIDIA confidential computing on H100 GPUs is a practical path.

Unlike ZKPs, which are computationally prohibitive for large models, TEEs are described as practical and deployable today.

KEY TAKEAWAYS

Close the Verification Gap with Hardware Attestation

The evidence points to a boundary: software probes can suggest anomalies, but cannot prove model identity in black-box LLM APIs.

01

Economic motive

Model substitution cuts GPU cost while preserving commercial pricing.

02

FP8 is subtle

Quantization barely affects many benchmark scores.

03

Log-probs drift

Inference nondeterminism defeats metadata fingerprinting.

04

Routing wins

Randomized substitution and benchmark evasion undermine tests.

05

TEEs verify

Hardware attestation is the viable path to verifiable APIs.

Community call:

Move LLM API integrity from probabilistic detection to attested execution.